

AGILE LAB ASSESSMENT-2

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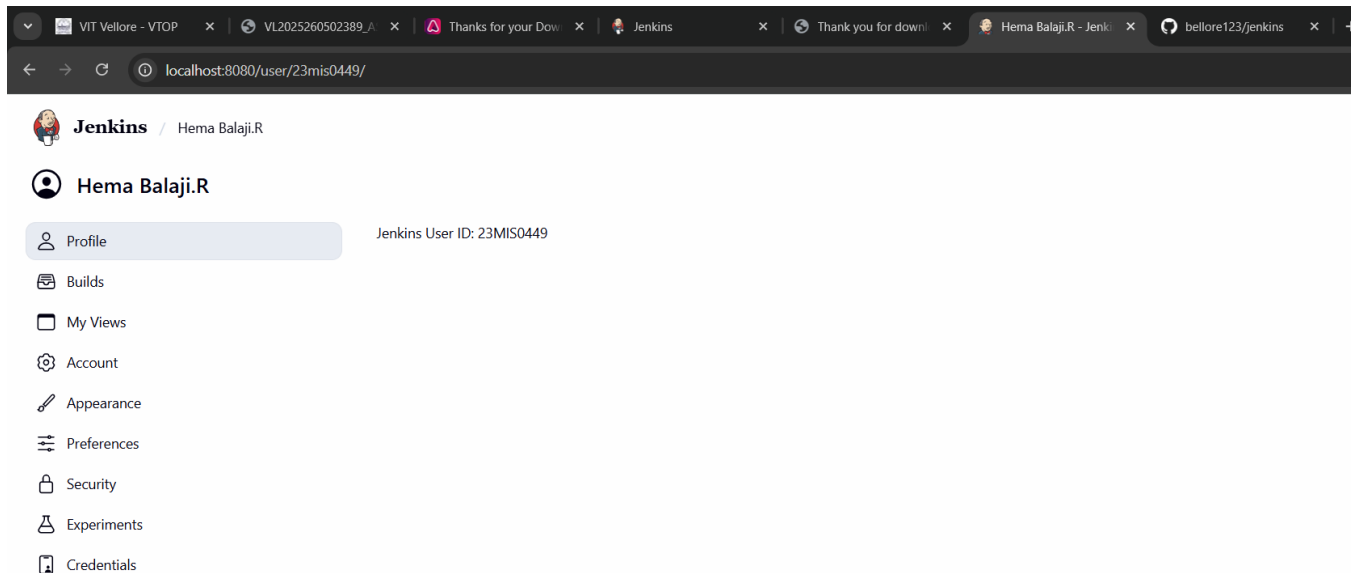
JENKINS

1. Learning Goal

Understand **what Jenkins is** and run a job successfully.

Stepwise Tasks

1. What is CI/CD? (Real-world example)
2. Install Jenkins (Windows / Linux)



3. Jenkins UI overview

4. Create **Freestyle Job**



New Item

Enter an item name

Select an item type



Pipeline

Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.



Freestyle project

Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.



Multi-configuration project

Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.



Folder

Creates a container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a filter, a folder creates a separate namespace, so you can have multiple things of the same name as long as they are in different folders.



Multibranch Pipeline

OK

5. Add simple build step:

- Windows: echo Hello Jenkins

Build Steps

Automate your build process with ordered tasks like code compilation, testing, and deployment.



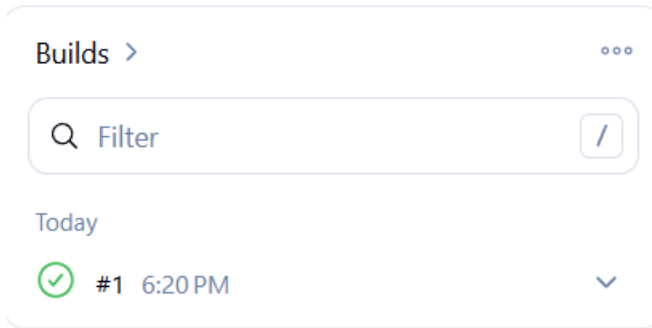
Execute Windows batch command ?

Command

[See the list of available environment variables](#)

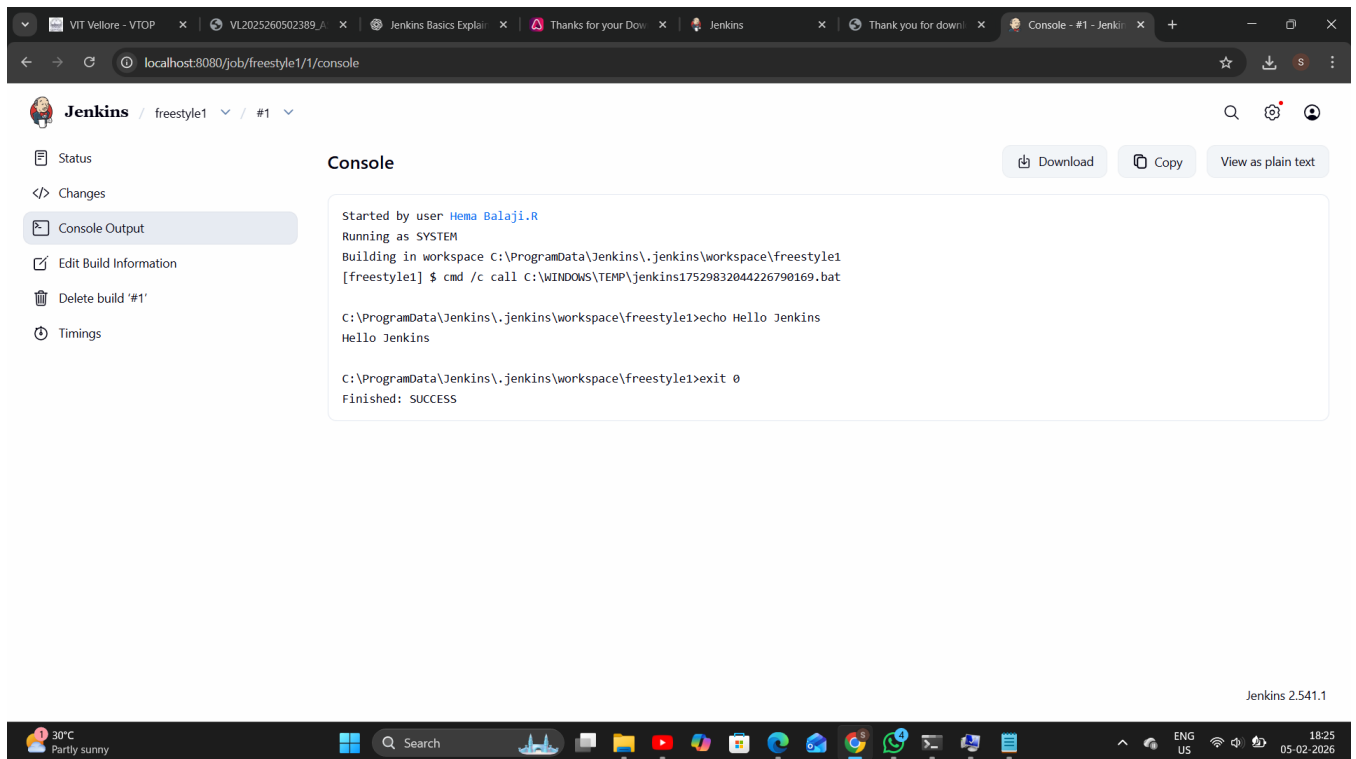
```
echo Hello Jenkins
```

6. Run job manually



7. View:

Console Output:



Build History:

Jenkins / freestyle1 / #1

Status **#1 (5 Feb 2026, 18:24:50)** [Add description](#) [Keep this build forever](#)

Changes

Console Output

Edit Build Information

Delete build '#1'

Timings

Started by user [Herna Balaji.R](#)

This run spent:

- 58 ms waiting;
- 0.5 sec build duration;
- 0.56 sec total from scheduled to completion.

No changes.

REST API Jenkins 2.541.1

Jenkins

+ New Item

Build History

Build Queue

No builds in the queue.

Build Executor Status

0/2

S	W	Name ↓	Last Success	Last Failure	Last Duration
✓	☀	freestyle1	8 min 10 sec #1	N/A	0.5 sec

Icons: S M L

2. Learning Goal

Work with **source code** and **build automation**

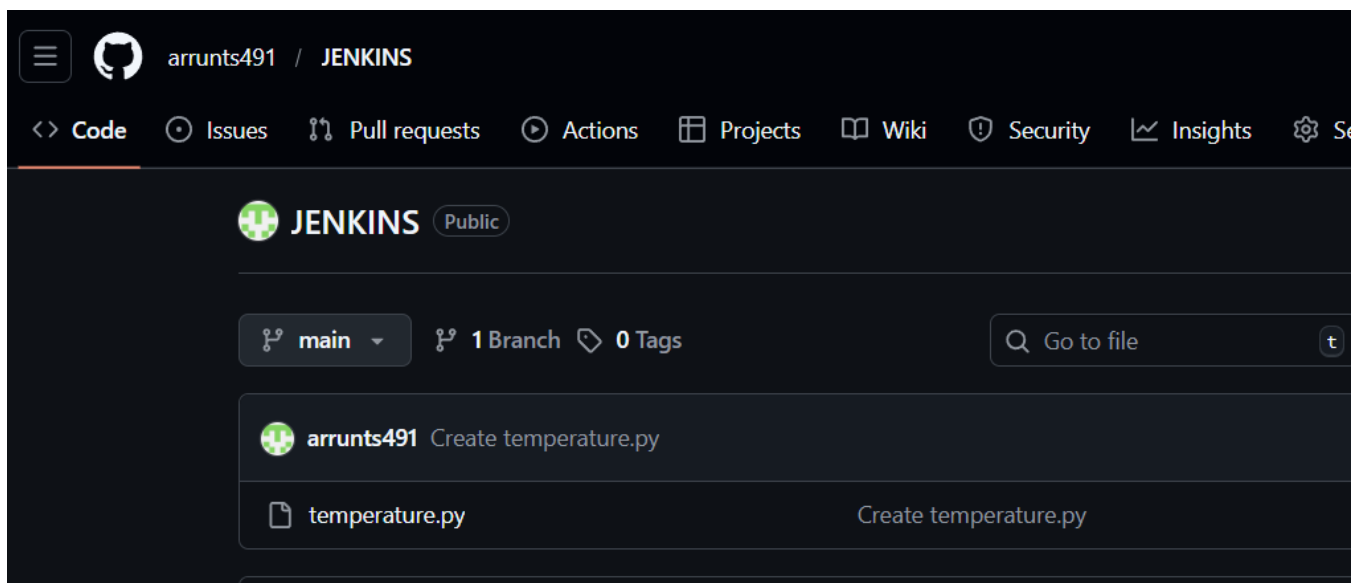
Stepwise Tasks

1. Install Git

```
C:\Users\sowmy>git --version
git version 2.52.0.windows.1
```

- 2.

3. Create GitHub repository



4. Configure **Git in Jenkins**

Git installations

Git

Name

AGILE

Path to Git executable ?

C:\Program Files\Git\bin\git.exe

5. Create Freestyle Job:

Source Code Management → Git:

Source Code Management

Connect and manage your code repository to automatically pull the latest code for your builds.

☐ None

☒ Git ?

Repositories ?

Repository URL ?

https://github.com/arrunts491/JENKINS.git

6. Add build step:

Windows batch / Shell

Build Steps

Automate your build process with ordered tasks like code compilation, testing, and deployment.

⋮ Execute Windows batch command ?

Command

See [the list of available environment variables](#)

```
python temperature.py
```

7. Poll SCM / Webhook trigger

Triggers

Set up automated actions that start your build based on specific events, like code changes or scheduled times.

- ☐ Trigger builds remotely (e.g., from scripts) ?
- ☐ Build after other projects are built ?
- ☐ Build periodically ?
- ☐ GitHub hook trigger for GITScm polling ?
- ☒ Poll SCM ?

Schedule ?

```
H/2 * * * *
```

8. Build automatically on code change

Builds > ...

/

Today

✓ #1 7:37 PM ▼

✓ AGILE_GITCLONE

Permalinks

- [Last build \(#1\), 5 min 27 sec ago](#)
- [Last stable build \(#1\), 5 min 27 sec ago](#)
- [Last successful build \(#1\), 5 min 27 sec ago](#)
- [Last completed build \(#1\), 5 min 27 sec ago](#)

Jenkins / freestyle1 / #6

Changes
Console Output
Edit Build Information
Delete build '#6'
Timings
Git Build Data
Previous Build

Started by user [Hema Balaji.R](#)

Running as SYSTEM

Building in workspace C:\ProgramData\Jenkins\jenkins\workspace\freestyle1

The recommended git tool is: NONE

No credentials specified

```

> git.exe rev-parse --resolve-git-dir C:\ProgramData\Jenkins\jenkins\workspace\freestyle1\.git # timeout=10
Fetching changes from the remote Git repository
> git.exe config remote.origin.url https://github.com/bellor123/jenkins.git # timeout=10
Fetching upstream changes from https://github.com/bellor123/jenkins.git
> git.exe --version # timeout=10
> git --version # 'git version 2.52.0.windows.1'
> git.exe fetch --tags --force --progress -- https://github.com/bellor123/jenkins.git +refs/heads/*:refs/remotes/origin/* # timeout=10
> git.exe rev-parse "refs/remotes/origin/main^{commit}" # timeout=10
Checking out Revision fb230ca28e917e533064726028ce2bcaec412519 (refs/remotes/origin/main)
> git.exe config core.sparsecheckout # timeout=10
> git.exe checkout -f fb230ca28e917e533064726028ce2bcaec412519 # timeout=10
Commit message: "Create hello.txt"
First time build. Skipping changelog.
[freestyle1] $ cmd /c call C:\WINDOWS\TEMP\jenkins12429647203877655636.bat

C:\ProgramData\Jenkins\jenkins\workspace\freestyle1>echo Hello Jenkins
Hello Jenkins

C:\ProgramData\Jenkins\jenkins\workspace\freestyle1>exit 0
Finished: SUCCESS
  
```

3. Learning Goal

Understand **job flow visually**

Stepwise Tasks

1. Create multiple Freestyle jobs:

- Job-A (Build)
- Job-B (Test)
- Job-C (Deploy)

		job A	N/A	N/A	N/A	
		job B	N/A	N/A	N/A	
		job C	N/A	N/A	N/A	

2. Use:

- Post-build action → Trigger other projects

FROM JOB-A-BUILD:



Jenkins

/ job B



/ Configure



echo testing

Advanced ▾

+ Add build step

Post-build Actions

Define what happens after a build completes, like sending notifications, archiving artifacts, or triggering other jobs.

Build other projects ?



Projects to build

job C

- ☒ Trigger only if build is stable
- ☐ Trigger even if the build is unstable
- ☐ Trigger even if the build fails

Save

Apply

FROM JOB-B-TEST:

3.View job chaining visually



Jenkins / job A



Status

</> Changes

Workspace

Build Now

Configure

Delete Project

Rename

Builds >

...



Filter



Today



#2 18:57



#1 18:54



job A

Downstream Projects



job B

Permalinks

- [Last build \(#1\), 2 min 16 sec ago](#)
- [Last stable build \(#1\), 2 min 16 sec ago](#)
- [Last successful build \(#1\), 2 min 16 sec ago](#)
- [Last completed build \(#1\), 2 min 16 sec ago](#)



Jenkins / job B



Status

</> Changes

Workspace



Build Now



Configure



Delete Project



Rename

Builds >

...



Filter



Today



#2 18:57



#1 18:55



job B

Upstream Projects



job A

Downstream Projects



job C

Permalinks

- Last build (#2), 2 min 1 sec ago
- Last stable build (#2), 2 min 1 sec ago
- Last successful build (#2), 2 min 1 sec ago
- Last completed build (#2), 2 min 1 sec ago

←

→

↻

🔍

localhost:8080/job/job%20C/

 **Jenkins** / job C

Status

Changes

Workspace

Build Now

Configure

Delete Project

Rename

🟢 job C

Upstream Projects

🟢 job B

Permalinks

- Last build (#2), 2 min 7 sec ago
- Last stable build (#2), 2 min 7 sec ago
- Last successful build (#2), 2 min 7 sec ago
- Last completed build (#2), 2 min 7 sec ago

Builds >

🔍 Filter

/

Today

🟢 #2 18:57

🟢 #1 18:55

4.Install Pipeline Stage View

 **Jenkins** / Manage Jenkins / Plugins

Plugins

Updates

Available plugins

Installed plugins

Advanced settings

🔍 pipeline stage view

/

Name 1	Health	Enabled
Pipeline: Stage View Plugin 2.39 Pipeline Stage View Plugin. Report an issue with this plugin	🟢 100	🔵

5.Observe stage-wise execution

4. Learning Goal

Use Jenkins Pipeline & Jenkinsfile

Stepwise Tasks

1. What is Jenkins Pipeline?
2. Scripted vs Declarative Pipeline
3. Create **Pipeline Job**

New Item

Enter an item name

Pipeline-Job

Select an item type



Pipeline

Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.

4. Write pipeline in Jenkins UI
5. Add stages:
 - Build
 - Test
 - Deploy
6. Introduce:
 - agent
 - stages
 - steps

Pipeline

Define your Pipeline using Groovy directly or pull it from source control.

Definition

Pipeline script

Script ?

```
1 pipeline {
2   agent any
3   stages {
4     stage('Build') {
5       steps {
6         echo 'Build Stage Running'
7       }
8     }
9     stage('Test') {
10      steps {
11        echo 'Test Stage Running'
12      }
13    }
14    stage('Deploy') {
15      steps {
```

```
14    stage('Deploy') {
15      steps {
16        echo 'Deploy Stage Running'
17      }
18    }
19  }
20 }
21
```


 **Jenkins** / pipeline

Status

</> Changes

▷ Build Now

⚙️ Configure

🗑️ Delete Pipeline

🔍 Full Stage View

📁 Stages

✎ Rename

❓ Pipeline Syntax

Builds >

Filter

/

Today

✓ #3 19:12

✓ #2 19:12

✓ #1 19:02

✓ pipeline

Stage View

Average stage times:
(full run time: ~2s)

	Build	Test	Deploy
#3 Feb 05 19:12 No Changes	144ms	121ms	134ms
#2 Feb 05 19:12 No Changes	153ms	135ms	149ms
#1 Feb 05 19:02 No Changes	244ms	123ms	125ms

Permalinks

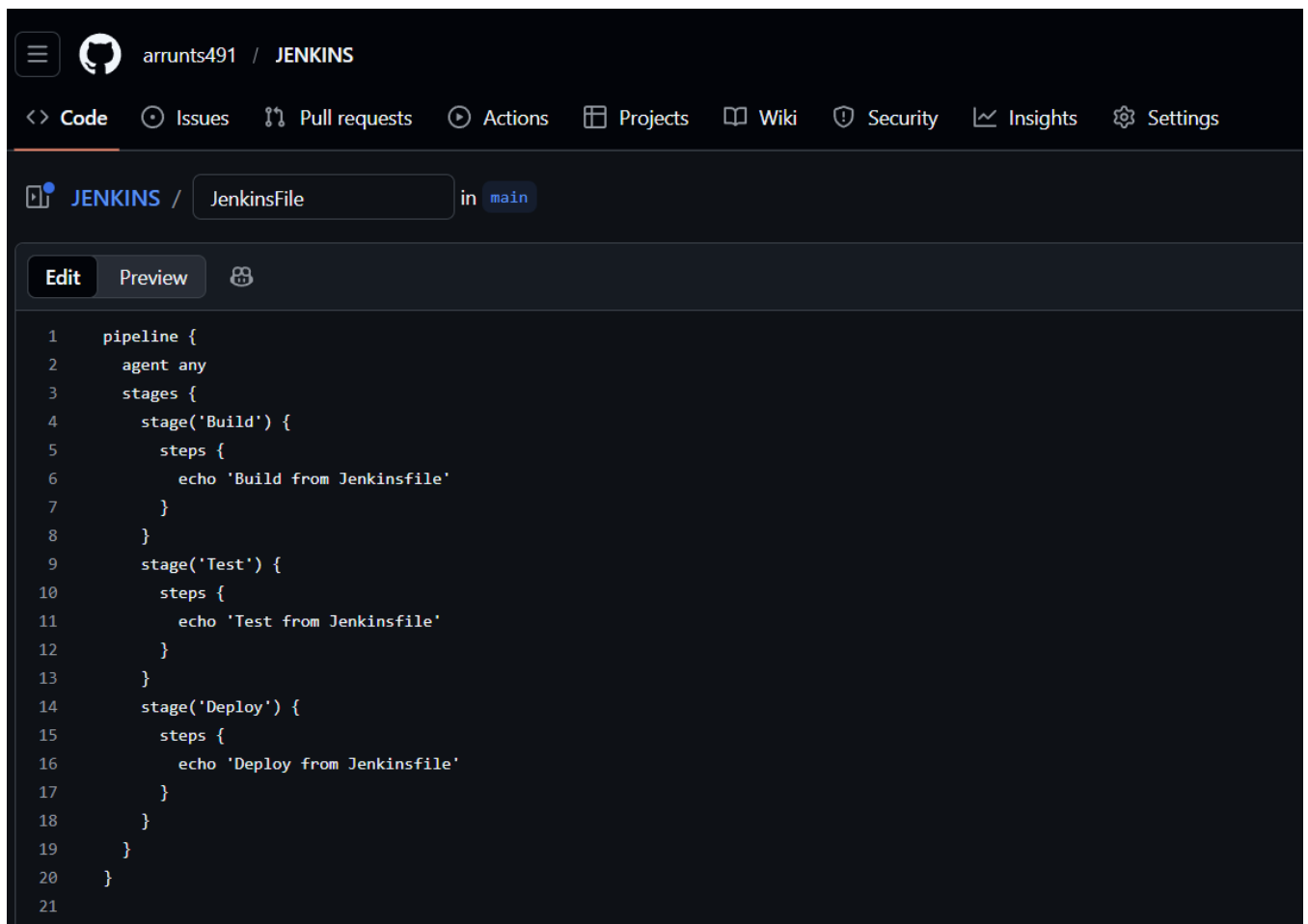
- [Last build \(#3\), 5 min 53 sec ago](#)
- [Last stable build \(#3\), 5 min 53 sec ago](#)
- [Last successful build \(#3\), 5 min 53 sec ago](#)
- [Last completed build \(#3\), 5 min 53 sec ago](#)

5. Learning Goal

Pipeline as code from repository

Stepwise Tasks

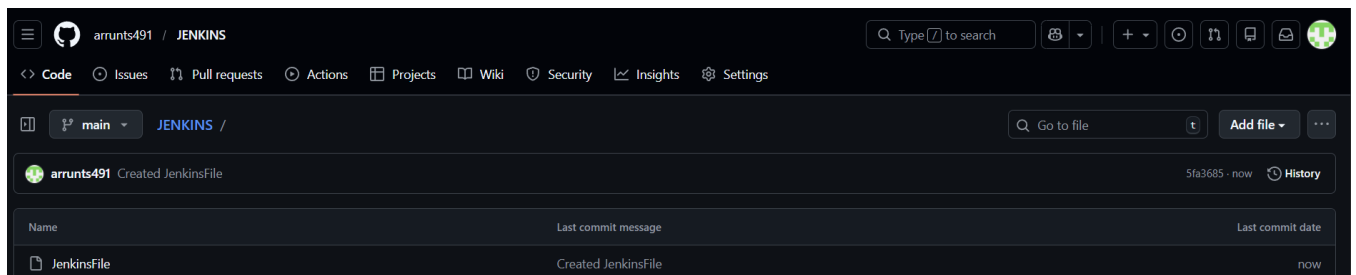
1.Create Jenkinsfile



The screenshot shows the GitHub web interface for a repository named 'JENKINS' by user 'arrunts491'. The 'Code' tab is selected, and the file 'JenkinsFile' is open in the 'main' branch. The file content is a Jenkins pipeline script with three stages: Build, Test, and Deploy. Each stage has an echo step. The interface includes navigation links like Issues, Pull requests, Actions, Projects, Wiki, Security, Insights, and Settings. At the top, there are buttons for 'Edit' and 'Preview'.

```
1 pipeline {
2   agent any
3   stages {
4     stage('Build') {
5       steps {
6         echo 'Build from Jenkinsfile'
7       }
8     }
9     stage('Test') {
10      steps {
11        echo 'Test from Jenkinsfile'
12      }
13    }
14    stage('Deploy') {
15      steps {
16        echo 'Deploy from Jenkinsfile'
17      }
18    }
19  }
20 }
21
```

2.Commit Jenkinsfile to GitHub



The screenshot shows the commit history for the 'JENKINS' repository. A single commit is listed, created by 'arrunts491' with the message 'Created JenkinsFile'. The commit hash is '5fa3685' and it was made 'now'. Below the commit list is a table with columns for Name, Last commit message, and Last commit date.

Name	Last commit message	Last commit date
JenkinsFile	Created JenkinsFile	now

3.Configure Pipeline job:

SCM → Git



Configure

General

Triggers

Pipeline

Advanced

Pipeline

Define your Pipeline using Groovy directly or pull it from source control.

Definition

Pipeline script from SCM

SCM ?

Git

Repositories ?

Repository URL ?

https://github.com/arrunts491/JENKINS.git

Credentials ?

- none -

Script Path ?

JenkinsFile

4.Trigger build automatically

First time build. Skipping changelog.

```
[Pipeline] }
[Pipeline] // stage
[Pipeline] withEnv
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Build)
[Pipeline] echo
Build from Jenkinsfile
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Test)
[Pipeline] echo
Test from Jenkinsfile
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Deploy)
[Pipeline] echo
Deploy from Jenkinsfile
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

 **Jenkins** / Git-Pipeline-Job

 Status

 Changes

 Build Now

 Configure

 Delete Pipeline

 Full Stage View

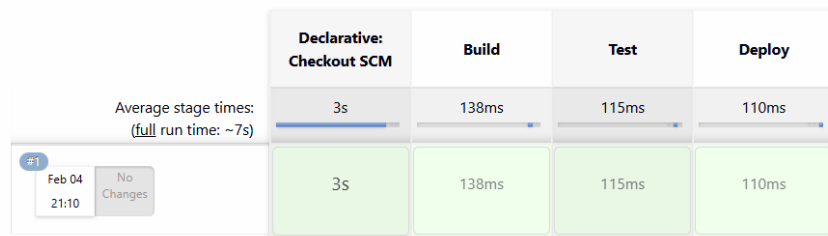
 Stages

 Rename

 Pipeline Syntax

 **Git-Pipeline-Job**

Stage View



Permalinks

- [Last build \(#1\), 8 min 46 sec ago](#)
- [Last stable build \(#1\), 8 min 46 sec ago](#)
- [Last successful build \(#1\), 8 min 46 sec ago](#)
- [Last completed build \(#1\), 8 min 46 sec ago](#)

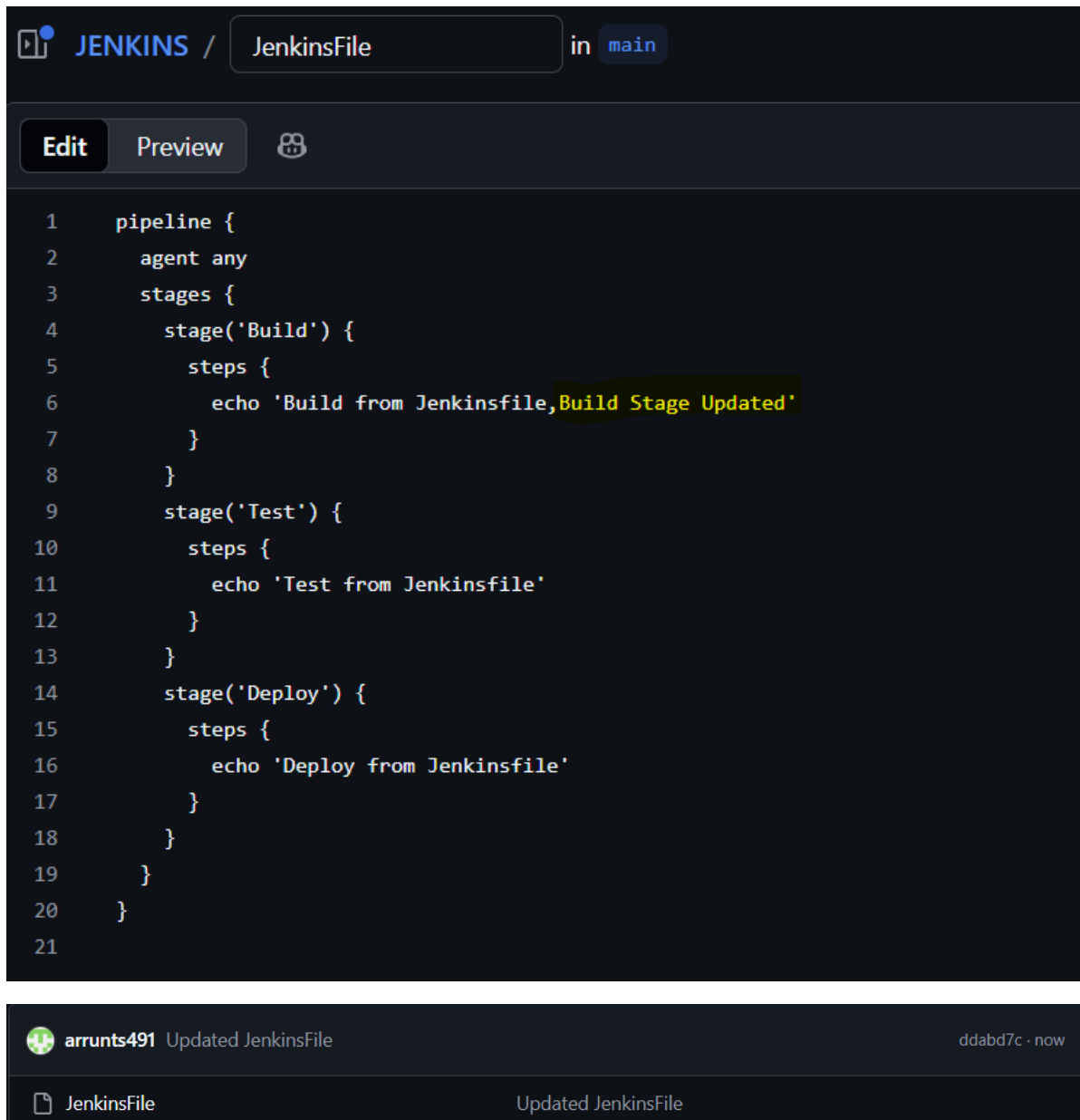
Builds >

 Filter

Today

 #1 9:10 PM

5.Modify Jenkinsfile and rebuild



The screenshot displays the Jenkins Jenkinsfile editor interface. At the top, the breadcrumb navigation shows 'JENKINS / JenkinsFile' with a dropdown menu currently set to 'main'. Below this, there are three tabs: 'Edit' (which is active), 'Preview', and a lock icon. The main area contains a YAML pipeline script with line numbers 1 through 21. The script defines a pipeline with an 'any' agent and three stages: 'Build', 'Test', and 'Deploy'. The 'Build' stage contains a single step that echoes 'Build from Jenkinsfile, Build Stage Updated', with the latter part of the string highlighted in yellow. The 'Test' stage echoes 'Test from Jenkinsfile', and the 'Deploy' stage echoes 'Deploy from Jenkinsfile'. At the bottom of the editor, a commit history bar shows a commit by 'arrunts491' titled 'Updated JenkinsFile' with the hash 'ddabd7c' and the message 'now'. Below this, a file list shows 'JenkinsFile' as the current file, with 'Updated JenkinsFile' and 'now' indicating the latest state.

```
1 pipeline {
2   agent any
3   stages {
4     stage('Build') {
5       steps {
6         echo 'Build from Jenkinsfile, Build Stage Updated'
7       }
8     }
9     stage('Test') {
10      steps {
11        echo 'Test from Jenkinsfile'
12      }
13    }
14    stage('Deploy') {
15      steps {
16        echo 'Deploy from Jenkinsfile'
17      }
18    }
19  }
20 }
21
```

arrunts491 Updated JenkinsFile ddabd7c · now 2 Commits

JenkinsFile Updated JenkinsFile now

UPDATED CONSOLE OUTPUT:

```

[Pipeline] }
[Pipeline] // stage
[Pipeline] withEnv
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Build)
[Pipeline] echo
Build from Jenkinsfile, Build Stage Updated
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Test)
[Pipeline] echo
Test from Jenkinsfile

```

6.View Stage View


Jenkins / Git-Pipeline-Job

Status

[Changes](#)
[Build Now](#)
[Configure](#)
[Delete Pipeline](#)
[Full Stage View](#)
[Stages](#)
[Rename](#)
[Pipeline Syntax](#)

Builds >

Today

#2 9:20 PM

#1 9:10 PM


Git-Pipeline-Job

Stage View

#2

Feb 04 21:20

1 commit

#1

Feb 04 21:10

No Changes

Average stage times:
(full run time: ~6s)

Declarative: Checkout SCM	Build	Test	Deploy
2s	125ms	104ms	97ms
2s	112ms	94ms	84ms
3s	138ms	115ms	110ms

Permalinks

- Last build (#2), 4 min 42 sec ago
- Last stable build (#2), 4 min 42 sec ago
- Last successful build (#2), 4 min 42 sec ago
- Last completed build (#2), 4 min 42 sec ago